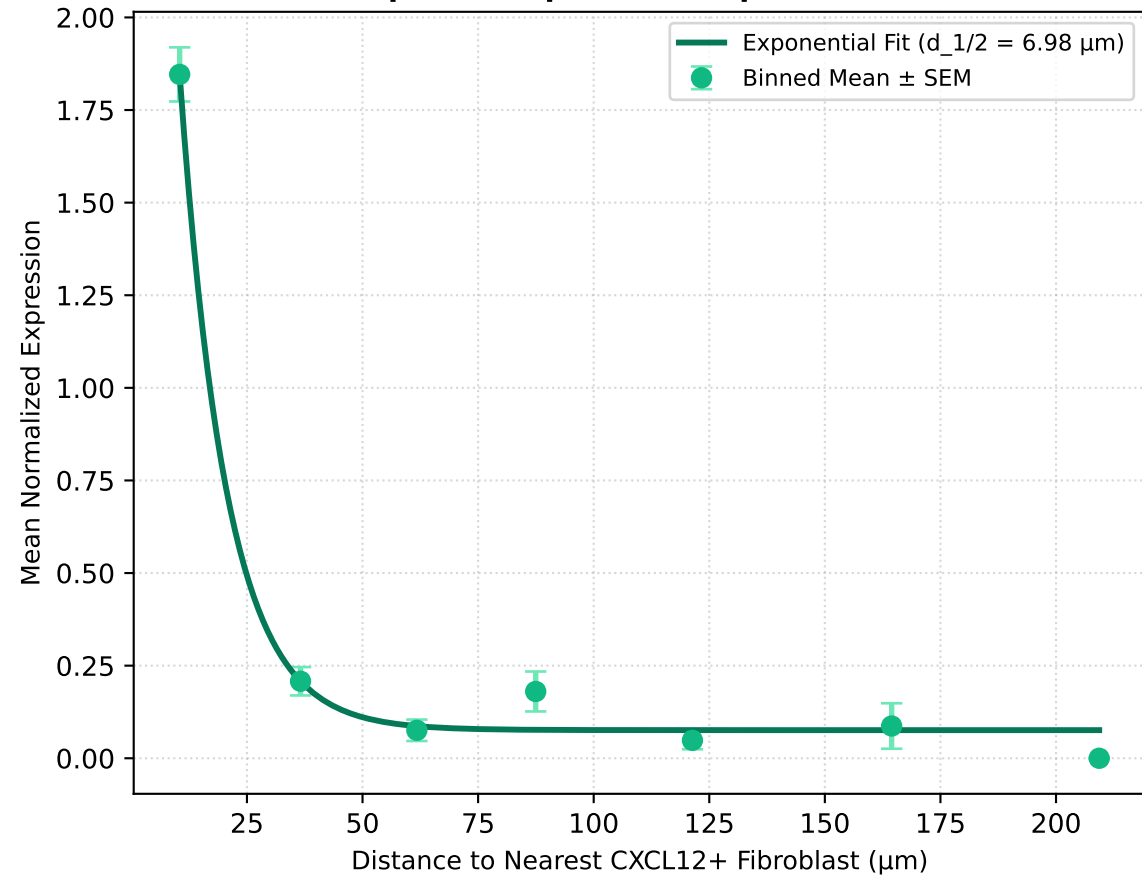
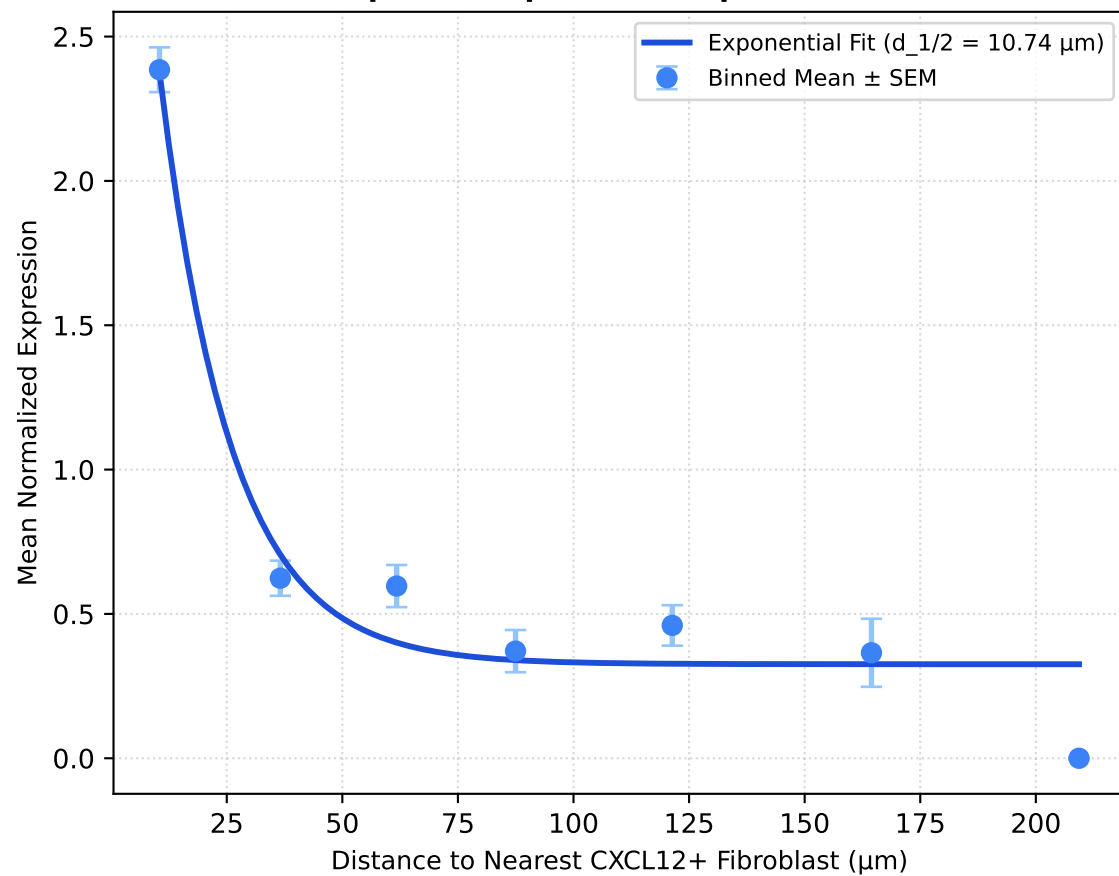


Single-Cell Dose-Response Kinetics: Quantitative Paracrine Extinction of CXCR4 Pathway Activation

S100A4 (Metastasis & EMT)
Spearman $\rho = -0.431$ ($p < 1e-15$)



MMP2 (Matrix Degradation)
Spearman $\rho = -0.447$ ($p < 1e-15$)



Kinase Pathway vs Proliferation Controls
Downstream Specificity across Distance Bins

